

Vishisht Rao

vsrao@cs.cmu.edu | vishisht.rao@gmail.com | (412)584-0828 | <https://vishisht-rao.github.io/> |
<https://www.linkedin.com/in/vishisht-rao/>

EDUCATION

Carnegie Mellon University, *PhD in Machine Learning*

Pittsburgh, PA, 08/2026 – Present

Selected Coursework: Intermediate Statistics (PhD)

Carnegie Mellon University, *MS in Machine Learning*; GPA: 3.89/4.00

Pittsburgh, PA, 08/2024 – 12/2025

Selected Coursework: Convex Optimization (PhD); Probabilistic Graphical Models (PhD); Deep Reinforcement Learning and Control (PhD)

PES University, *BTech in Computer Science and Engineering*; GPA: 9.08/10.0

Bangalore, India, 08/2019 – 06/2023

Recipient of the MRD Scholarship, awarded to the top 20% of students, in 5 out of 6 possible semesters

PUBLICATIONS

- [1] **Vishisht Rao**, Justin Payan, Andrew McCallum, Nihar B. Shah, “ML Researchers Support Openness in Peer Review But Are Concerned About Resubmission Bias”. Preprint.
Link: <https://arxiv.org/abs/2511.23439>
Code: <https://github.com/justinpayan/OpenReviewAnalysis>
- [2] Sarina Xi, **Vishisht Rao**, Justin Payan, Nihar B. Shah, “FLAWS: A Benchmark for Error Identification and Localization in Scientific Papers”. Preprint.
Link: <https://arxiv.org/abs/2511.21843>
Code: <https://github.com/xasayi/FLAWS>
- [3] **Vishisht Rao**, Aounon Kumar, Himabindu Lakkaraju, Nihar B. Shah, “Detecting LLM-Generated Peer Reviews”. PLOS ONE. Podium presentation at the *Tenth International Congress of Peer Review and Scientific Publication*.
Link: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0331871>
Code: <https://github.com/Vishisht-rao/detecting-llm-written-reviews>
- [4] Mayank Ratan Bhardwaj, **Vishisht Rao**, Bazil Ahmed, Kartik Sagar, Y. Narahari, “Deep Learning Based Auction Design for Sale of Agricultural Produce through Farmer Collectives to Maximize Nash Social Welfare”. Preprint.
Link: <https://arxiv.org/abs/2512.22039>
- [5] V. Udaya Sankar, **Vishisht Rao**, Y. Narahari, “Deep Learning Meets Mechanism Design: Key Results and Some Novel Applications”. Preprint.
Link: <https://arxiv.org/abs/2401.05683>
- [6] Prajwal Gupta, **Vishisht Rao**, Rithvik Ganesh, R. L. Sameeksha, V. R. Badri Prasad, “SaliCLIP: Generating Speech-Conditioned Images by Aligning Speech With CLIP Latents” 2023 8th International Conference on Image, Vision and Computing (ICIVC), Dalian, China, 2023, pp. 231-238
Link: <https://ieeexplore.ieee.org/abstract/document/10270664>
Code: <https://github.com/Vishisht-rao/SaliCLIP>

RESEARCH EXPERIENCE

Carnegie Mellon University (CMU), *Research Assistant*

Pittsburgh, PA, 09/2024 – 12/2025

• Detecting LLM-Generated Peer Reviews

- Built a three-component framework for detecting LLM-generated reviews – *watermarking* strategy that stochastically selects specific phrases to serve as detectable signals in LLM-generated reviews – *indirect prompt injection* techniques to embed instructions into manuscript PDFs – *statistical detection method* that identifies embedded watermarks across multiple reviews with formal FWER control and higher power than standard corrections; flagged LLM reviews with high 100% TPR at zero false flags under strict FWER bound of 0.01.
- Watermark persisted across 100 generations across several frontier models.
- Sparked key discussion in *The Workshop on Practical Improvements to Peer Review, AAMAS 2025*, presented as a podium talk at *The 10th International Congress on Peer Review and Scientific Publication*, presented as a poster at the *AI-SDM Workshop on Human-AI Complementarity for Decision Making (Carnegie Mellon University)*, and published in *PLOS ONE*.

• Benchmarking LLMs for Error Detection in Scientific Papers

- Developing a benchmark dataset consisting of scientific papers with subtle errors introduced to evaluate the ability of LLMs to identify such errors
- Designed a multi-stage pipeline to identify key claims made in scientific papers and introduce a subtle modification in the paper to invalidate these claims using LLMs
- Showed through pilot tests that errors inserted have < 40% detection across SOTA LLMs

• Effect of Open vs Closed Reviewing Systems on Quality of Peer Reviews

- Designed a human-subject study and conducted a survey with over 2,300 responses from ICLR 2025 participants to to inform conference policy on when open reviewing is actually beneficial.

- Built an LLM-based evaluator to score real review texts from the two settings on metrics such as substantiation, correctness, and completeness.

Indian Institute of Science (IISc), *Research Intern, Research Engineer*

Bangalore, India, 12/2022 – 07/2024

- **Deep Learning Based Auction Mechanisms**

- Designed data-driven auctions using neural networks for the procurement of supply materials, and the sale of produce, in the agricultural domain
- Trained allocation and payment networks for volume-discount and combinatorial settings, optimizing a composite objective to minimize the violation of Nash social welfare, individual rationality, incentive compatibility, envy-freeness, and practical business constraints
- Worked closely with two agricultural districts in Karnataka, India to implement relevant state-of-the-art solutions, which won the *Elets Technology Excellence Award 2024* in the Public Sector Initiatives for Social Good category

- **Deep Learning Meets Mechanism Design - A Survey**

- Surveyed and synthesized DL-based approaches to auction/mechanism design; mapped literature to core properties (IC/IR, revenue, welfare, fairness, business constraints)
- Proposed a clear taxonomy and design checklist for practitioners; contrasted characterization-based vs. characterization-free methods and their regret/complexity profiles

ACADEMIC PROJECTS

- **Generating Speech-Conditioned Images – Undergraduate Capstone Project**

01/2022 – 12/2022

- Learned a shared embedding space by pulling paired speech-image samples together and pushing negatives apart; aligned speech encoders to CLIP image features
- Enabled direct speech-conditioned image generation (no text intermediary) and efficient cross-modal retrieval
- Reported state-of-the-art retrieval metrics and proposed a novel cross-modal classification trick with strong accuracy

TEACHING EXPERIENCE

TA – Advanced Introduction to Machine Learning (PhD) [10-715], CMU

Aug. 2025 – Dec. 2025

- Designed homeworks on SVMs, optimization, Kernel methods, neural networks; conducted a recitation on tail bounds
- Graded homeworks, midterms, final exams, and held office hours

TA – Game Theory Course [E1 254], IISc

Jan. 2024 – May 2024

- Conducted problem solving sessions on strategic form games, Nash equilibria, and two-player zero-sum games
- Organized and scheduled project presentations of over 100 students, assisted professors in reviewing all projects and presentations along with other TAs
- Held office hours, graded midterm papers, invigilated exams, and tabulated results of exams, assignments, projects

WORK EXPERIENCE

Bosch Global Software Technologies, *Summer Intern*

Bangalore, India, 06/2022 – 07/2022

- Built applications that provided proof of concepts for aggregation and visualization techniques for location-based data using FastAPI, React and S2Cell, which contributed to the Data Engineering team that plays an integral role in developing autonomous driving solutions

LEADERSHIP & PROFESSIONAL SERVICE

- **Reviewer:** Tenth International Congress on Peer Review and Scientific Publication

- **Volunteer Technologist – Project StepOne**

Sept. 2022 – Aug. 2024

- Developed a helpline platform to address mental health-related issues that was adopted by 21 state governments.
- Designed Finite State Machines to automatically connect individuals to healthcare professionals based on their needs.

- **Local Organizing Committee Member – GAMEARTS (Game Theory, Mechanism Design, and Artificial Intelligence)**

July 2024

- Led the organization of the symposium involving over 450 attendees from multiple countries, held at IISc.
- Planned logistics for talks, guest travel, accommodation, catering, funding, and felicitation ceremonies.

- **Student Ambassador – 16th International Conference on Educational Data Mining**

July 2023

- Played a pivotal role in the organization and coordination of events for over 219 participants from 20 different countries.
- Organized logistics for distinguished guests; coordinated with the technical support team for audio visuals, poster displays, and demo setups; planned and coordinated tasks for all Student Ambassadors.

- **Admissions Committee**

- Served on the admissions committee for the Master of Science in Machine Learning program at CMU (Fall 2025 intake)
- Reviewed more than 200 applications followed by multiple calibration rounds

- **Interviewer:** Interviewed 10 candidates for Teaching Assistant positions for Spring 2026, in the Machine Learning Department at CMU